Perceptual Image Denoising using Multi-Scale U-Net with Model Optimization

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***Abstract*—Image denoising is a fundamental problem in computer vision, particularly for real-world noisy images captured by mobile devices. Traditional filtering methods often fail to preserve fine details while removing noise. In this work, we propose a deep learning-based approach using a U-Net architecture enhanced with perceptual loss and multi-scale training. We further explore model optimization techniques including mixed precision training and structured pruning to improve efficiency. Experimental results on the SIDD dataset demonstrate that the proposed model significantly outperforms classical filtering methods, achieving a PSNR of 38.39 dB and strong perceptual quality. Additionally, we analyze the trade-off between model compression and performance through pruning experiments.**

I. Introduction

Image denoising is a fundamental problem in computer vision, aiming to recover clean images from noisy observations. Noise in images can arise from various sources such as low-light conditions, sensor limitations, and electronic interference, especially in mobile photography. Effective denoising is critical for downstream tasks including object detection, medical imaging, and image enhancement.

Traditional image denoising techniques, such as Gaussian filtering, median filtering, and bilateral filtering, are widely used due to their simplicity and computational efficiency. However, these methods often suffer from a trade-off between noise removal and detail preservation, frequently resulting in over-smoothed outputs and loss of fine textures. Classical methods such as BM3D [1] have shown strong performance but are limited in handling complex real-world noise.

Recent advances in deep learning have significantly improved the performance of image denoising systems. Convolutional neural networks (CNNs), such as DnCNN [2], learn noise distributions directly from data and achieve superior results compared to classical methods. Encoder-decoder architectures like U-Net [3] further enhance performance by combining hierarchical feature extraction with skip connections that preserve spatial information.

In addition to pixel-wise reconstruction losses, perceptual loss functions have been introduced to improve visual quality. These losses, computed using deep feature representations, better capture human perceptual similarity compared to traditional L1 or L2 losses [4]. This is particularly important for real-world datasets where preserving texture and structural details is essential. Residual learning techniques [5] have also influenced modern denoising architectures.

Despite these advancements, two key challenges remain. First, models trained on fixed-resolution inputs often struggle to generalize across varying image sizes and noise patterns. Second, high-performing deep models are computationally expensive, limiting their deployment on resource-constrained devices.

In addition to improving denoising quality, recent research has also focused on balancing model performance with computational efficiency. This is particularly important for deployment in real-world applications such as mobile photography and embedded systems, where resource constraints are significant. Motivated by this, our work not only explores improvements in denoising performance through architectural and training enhancements but also investigates lightweight optimization techniques such as pruning and efficient inference strategies. This dual focus enables a more comprehensive understanding of the trade-offs between accuracy, perceptual quality, and deployment feasibility.

A. Motivation

The motivation of this work is twofold. First, we aim to improve denoising performance on real-world noisy images by incorporating perceptual loss and multi-scale training. Multi-scale learning enables the model to capture noise characteristics across different spatial resolutions, leading to better generalization.

Second, we aim to address the efficiency-performance trade-off by exploring model optimization techniques such as mixed precision training and weight pruning. These techniques can significantly reduce computational cost and memory usage while maintaining acceptable performance.

B. Related Works

In addition to convolutional architectures, recent approaches have explored transformer-based models such as RED-Net [6] for image restoration tasks. These models leverage self-attention mechanisms to capture long-range dependencies, which can be beneficial for modeling complex noise patterns. However, they often come with increased computational cost and require large-scale training data. In contrast, convolutional architectures such as U-Net remain competitive due to their efficiency and strong performance on moderate-sized datasets. This motivates our choice of architecture, which balances performance and computational practicality.

C. Contributions

The main contributions of this work are as follows:

- We propose a U-Net based image denoising framework enhanced with perceptual loss for improved visual quality.
- We introduce multi-scale training to improve generalization across varying image resolutions.
- We apply mixed precision training to accelerate training and reduce memory consumption.
- We systematically analyze the impact of model pruning and demonstrate the trade-off between efficiency and performance.

D. Expected Outcomes

We expect the proposed approach to outperform traditional filtering methods and baseline CNN models in terms of both quantitative metrics (PSNR, SSIM) and perceptual quality (LPIPS). Additionally, we expect moderate pruning levels to retain most of the model performance while enabling more efficient deployment.

II. METHODOLOGY

A. Dataset

We conduct our experiments on the Smartphone Image Denoising Dataset (SIDD), which contains real-world noisy images paired with corresponding clean ground truth [7] images. Unlike synthetic datasets, SIDD captures realistic noise patterns arising from camera sensors under varying lighting conditions. The dataset is organized into multiple scenes, each containing noisy-clean image pairs. We split the dataset into training, validation, and test sets for robust evaluation.

B. Baseline Methods

To establish a reference, we evaluate classical denoising techniques including Gaussian, median, and bilateral filtering. These traditional approaches are widely used due to their simplicity, but they often suffer from over-smoothing and loss of fine details, motivating the use of learning-based methods.

C. Network Architecture

Our model is based on the U-Net architecture [3], a widely adopted encoder-decoder framework originally proposed for biomedical image segmentation. U-Net has also been successfully applied to image restoration tasks due to its ability to capture both local and global context.

The encoder-decoder structure of U-Net enables hierarchical feature extraction, where lower layers capture fine-grained details and deeper layers capture more abstract representations. Skip connections play a crucial role in preserving spatial information, allowing the network to recover sharp details in the reconstructed image. This is particularly important for denoising tasks, where excessive smoothing can lead to loss of texture and edges.

The encoder progressively downsamples the input image to extract hierarchical features, while the decoder reconstructs the clean image. Skip connections between corresponding encoder and decoder layers help preserve spatial information and fine textures, which are critical for image denoising.

Our design is further inspired by CNN-based denoising models such as DnCNN [2] and FFDNet [8], which demonstrate the effectiveness of deep convolutional architectures in modeling complex noise distributions.

D. Loss Function

We use a combination of pixel-wise L1 loss and perceptual loss. While L1 loss ensures accurate reconstruction at the pixel level, perceptual loss captures high-level semantic similarity between images. The perceptual loss is computed using deep feature representations as proposed in [4], which has been shown to correlate well with human visual perception.

E. Single-Scale Training

As an initial baseline, the model is trained using fixed-size patches (e.g., 128×128) randomly cropped from the training images. This setup allows efficient training while maintaining sufficient spatial context for learning noise patterns.

F. Multi-Scale Training

To improve generalization, we extend the training process to multiple resolutions. Specifically, patches of varying sizes (e.g., 64×64, 96×96, 128×128, and 192×192) are used during training. For each epoch, a patch size is randomly selected, and all training samples are generated accordingly. This multi-scale strategy enables the model to learn noise characteristics across different spatial resolutions.

G. Efficient Training on GPU

Training high-resolution images requires significant computational resources. To address this, we adopt several optimization strategies:

- **Patch-based training:** Instead of full-resolution images, smaller patches are used to reduce memory consumption.
- **Mixed precision training:** Automatic mixed precision (AMP) is used to accelerate training and reduce GPU memory usage.
- **Tiled inference:** During evaluation, large images are processed in overlapping tiles to avoid memory limitations. Padding is applied to ensure compatibility with the network architecture.

To further stabilize training, careful consideration was given to learning rate selection and batch size. The use of adaptive optimization methods allows the model to converge efficiently, while mixed precision training reduces memory usage and accelerates computation. These design choices enable effective utilization of available GPU resources, making large-scale training feasible.

H. Model Pruning

To improve computational efficiency, we apply unstructured weight pruning to the trained model. A fraction of weights with small magnitudes is removed, resulting in a sparse model. We experiment with multiple pruning levels (15% to 40%) to analyze the trade-off between model compression and denoising performance.

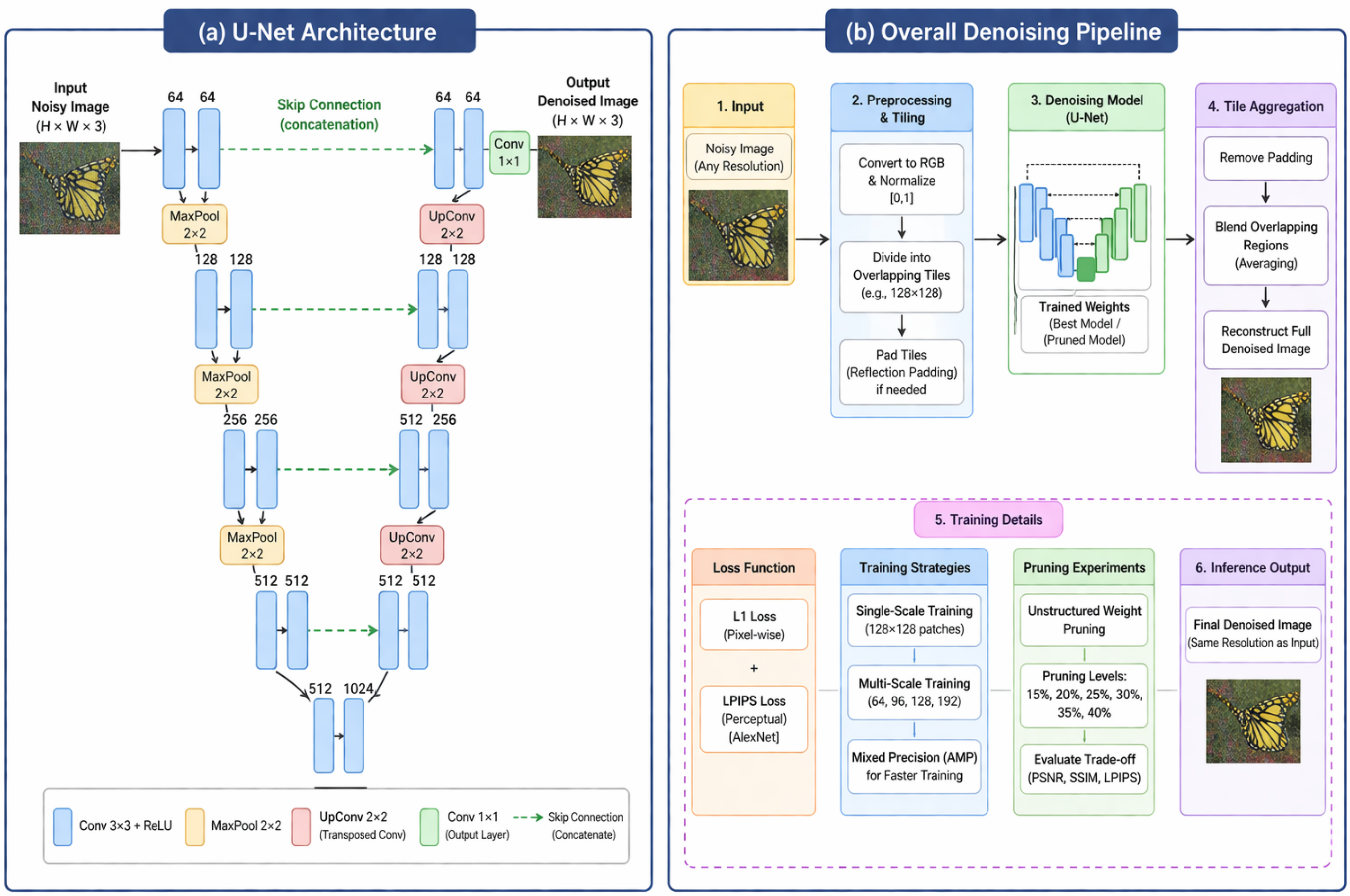

Fig. 1. Overview of the proposed image denoising framework. The left panel illustrates the U-Net architecture used for denoising, inspired by [3]. The right panel shows the complete pipeline, including preprocessing, tiling, model inference, and post-processing for reconstructing the final denoised image.

I. Experimental Setup

The model is trained using the Adam optimizer with a learning rate of 10^{-4}. Training is performed on a GPU using mixed precision. Performance is evaluated on the test set using standard image quality metrics. Data augmentation through random cropping further improves model robustness and prevents overfitting.

J. Evaluation Metrics

We evaluate the performance using:

- **PSNR:** Measures pixel-level reconstruction fidelity.
- **SSIM:** Captures structural similarity between images.
- **LPIPS:** Evaluates perceptual similarity using deep feature representations [4].

III. Results and Discussion

A. Comparison with Baseline Methods

We first compare our deep learning model with traditional image denoising techniques. Classical filters are computationally efficient but often fail to preserve fine details.

The results show that the U-Net model significantly outperforms traditional filtering techniques. While classical methods tend to smooth out noise along with important details, the deep learning model preserves textures and structures more effectively.

Method	PSNR	SSIM	LPIPS
Gaussian Filter	32.33	0.73	–
Median Filter	30.95	0.67	–
Bilateral Filter	34.36	0.82	–
U-Net (Single Scale)	**37.39**	**0.925**	0.112

TABLE I
Comparison with Classical Denoising Methods

B. Impact of Multi-Scale Training

We evaluate the effect of multi-scale training compared to single-scale training.

Model	PSNR	SSIM	LPIPS
U-Net (Single Scale)	37.39	0.925	0.112
U-Net (Multi-Scale)	**38.39**	**0.928**	**0.106**

TABLE II
Effect of Multi-Scale Training

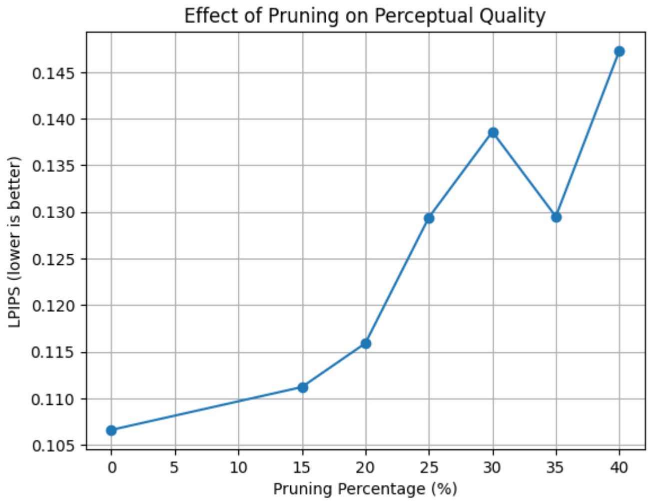

Fig. 2. LPIPS variation across different pruning levels. Lower values indicate better perceptual quality. The plot shows that perceptual quality degrades gradually with increased pruning.

Multi-scale training improves performance across all metrics. The model benefits from exposure to varying resolutions, enabling better generalization to diverse image sizes and noise patterns.

C. Effect of Model Pruning

To analyze the trade-off between efficiency and performance, we evaluate the model under different pruning levels.

Pruning (%)	PSNR	SSIM	LPIPS
0 (Full Model)	38.39	0.928	0.106
15	35.83	0.926	0.111
20	35.21	0.925	0.116
25	31.46	0.914	0.129
30	29.67	0.910	0.139
35	29.91	0.908	0.130
40	28.49	0.902	0.147

TABLE III
PERFORMANCE UNDER DIFFERENT PRUNING LEVELS

The results indicate that moderate pruning (15–20%) maintains acceptable performance with only a small drop in PSNR and SSIM. However, aggressive pruning beyond 25% leads to a sharp degradation in performance, indicating that the U-Net architecture is sensitive to excessive weight removal.

D. Inference Performance and Trade-offs

For inference, the multi-scale trained model (without pruning) achieves the best performance in terms of both quantitative and perceptual metrics. This model is suitable for high-quality applications where accuracy is the primary concern.

On the other hand, pruned models offer a trade-off between efficiency and performance. The 15% pruned model maintains high structural similarity and perceptual quality while reducing model complexity, making it suitable for deployment on resource-constrained devices. Higher pruning levels further reduce computational cost but result in significant degradation in output quality.

Tiled inference is employed to handle high-resolution images that may not fit entirely in GPU memory. By processing overlapping patches and aggregating the outputs, the model can scale to larger images without significant loss of quality. This approach also ensures that boundary artifacts are minimized, leading to more consistent predictions across the image.

E. Qualitative Results

In addition to quantitative evaluation, we visually inspect the denoised outputs produced by the proposed model. The results demonstrate that the model effectively removes noise while preserving fine textures and structural details. Compared to traditional filtering methods, which often produce over-smoothed outputs, the proposed approach maintains sharper edges and more natural image appearance.

We also observe that multi-scale training improves robustness across different image sizes, producing consistent results on both small and large images. The pruned models retain visually acceptable quality at moderate pruning levels (15–20%), although slight degradation becomes noticeable at higher pruning levels.

F. Discussion

The experimental results demonstrate that deep learning-based approaches significantly outperform traditional denoising methods. Multi-scale training further enhances performance by improving robustness across different resolutions.

The pruning experiments reveal an important insight: while moderate pruning can achieve a good balance between efficiency and performance, aggressive pruning disrupts the learned feature representations, leading to substantial quality loss. This highlights the importance of carefully selecting pruning levels for deployment scenarios.

Overall, the proposed approach achieves strong performance on real-world noisy images and provides a flexible framework that balances accuracy and efficiency.

IV. CONCLUSION

In this work, we presented a deep learning-based image denoising framework using a U-Net architecture enhanced with perceptual loss and multi-scale training. The proposed approach significantly outperforms traditional filtering methods, achieving a PSNR of 38.39 dB along with strong structural and perceptual similarity metrics.

Beyond quantitative improvements, the model demonstrates strong qualitative performance by effectively removing noise while preserving fine textures and edges. This is particularly important for real-world datasets such as SIDD, where noise characteristics are complex and cannot be easily modeled using traditional approaches.

Multi-scale training was shown to play a critical role in improving generalization across varying image resolutions. By exposing the model to patches of different sizes, it learns more robust representations of noise patterns. Additionally, mixed precision training enabled efficient utilization of GPU resources, reducing training time and memory consumption without compromising performance.

Furthermore, the integration of perceptual loss contributes to improved visual quality by aligning the model outputs with human perception. This ensures that the denoised images are not only numerically accurate but also visually realistic.

Overall, the proposed pipeline provides a robust and efficient solution for real-world image denoising tasks, particularly in applications involving mobile photography and low-light imaging scenarios.

V. Limitations

Despite strong performance, the proposed approach has several limitations that should be considered.

First, the model relies on supervised learning and requires paired noisy-clean datasets for training. Such datasets are expensive to collect and may not always be available in practical applications. This limits the scalability of the approach to new domains where ground truth clean images are unavailable.

Second, while tiled inference enables processing of high-resolution images, it introduces additional computational overhead and increases inference time. Although overlap and padding strategies mitigate boundary artifacts, minor inconsistencies may still arise at tile boundaries in some cases.

Third, the U-Net architecture, although effective, is relatively heavy in terms of computational and memory requirements. This can make deployment on edge devices or real-time systems challenging without further optimization.

Additionally, the pruning experiments reveal that the model is sensitive to aggressive weight removal. While moderate pruning (15–20%) maintains acceptable performance, higher pruning levels lead to significant degradation in both quantitative and perceptual metrics. This indicates that the learned feature representations are highly dependent on network capacity.

Finally, the model is trained specifically for denoising and may not generalize well to other restoration tasks such as deblurring or super-resolution without architectural modifications.

VI. Future Work

Future work can explore several directions to further improve the proposed approach.

One promising direction is the use of unsupervised or self-supervised learning techniques, which can reduce dependence on paired datasets. Methods such as noise-to-noise or noise-to-void training could enable learning directly from noisy data.

Another direction is the exploration of advanced architectures, including attention-based models and transformer-based networks. These models have shown strong performance in capturing long-range dependencies and may further improve denoising quality.

Model optimization techniques can also be extended. Structured pruning, quantization, and knowledge distillation could be investigated to achieve better compression while maintaining performance. Additionally, deployment-specific optimizations such as TensorRT or hardware-aware training can further improve inference speed.

Real-time inference remains an important challenge. Future work could focus on designing lightweight architectures that achieve a better balance between speed and accuracy, making them suitable for mobile and embedded applications.

Finally, extending the model to related tasks such as video denoising, low-light enhancement, and joint restoration tasks presents an exciting opportunity. Incorporating temporal information in video or multi-frame settings could significantly enhance denoising performance.

References

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Appendix A
Qualitative Results

This appendix presents qualitative comparisons between the input noisy images and the outputs produced by the proposed U-Net-based denoising model. The results demonstrate subtle yet consistent noise reduction, particularly in low-light and smooth regions, while preserving structural details.

To better visualize the results, we extract moderately sized patches from three regions of the image: center $(600, 300, 1400, 1000)$, left $(300, 200, 1100, 900)$, and right $(900, 300, 1700, 1000)$. These regions capture a mix of smooth areas and edges, enabling clearer observation of noise reduction while maintaining sufficient context.

Appendix B
Project Repository

The complete implementation is available at:
https://github.com/rano-dipa/image-denoising-unet

The repository includes all components necessary to reproduce the results presented in this work, including training scripts, evaluation pipelines, model checkpoints, and SLURM job configurations. It also provides example scripts for inference on unseen images, enabling straightforward application of the proposed model to real-world scenarios.

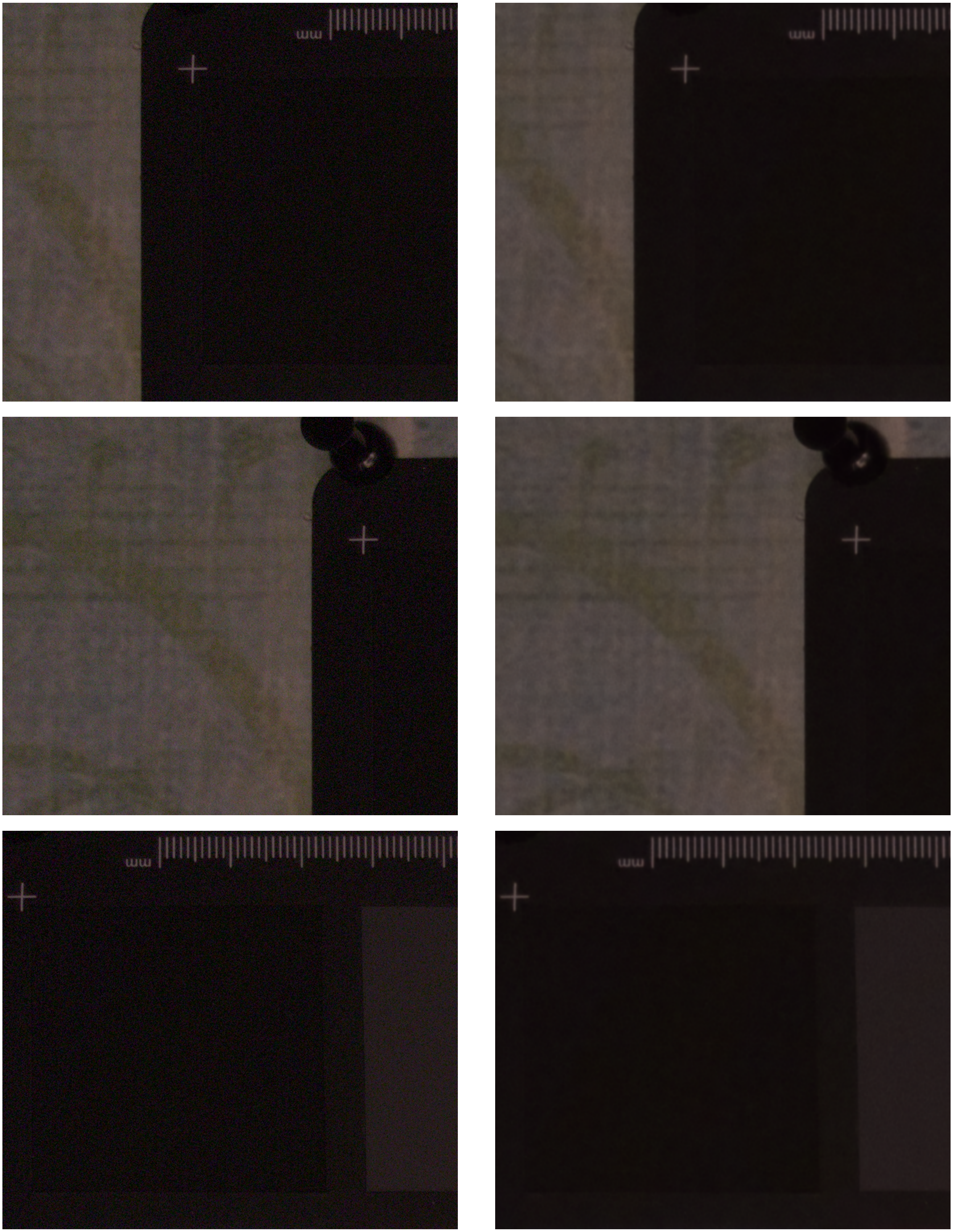

Fig. 3. Qualitative comparison of denoising results. Left: original noisy crops. Right: outputs from the proposed model. The model reduces noise while preserving structural details.